

Chapter 21 Hexapods

Subphylum Hexapoda

- Characteristics:
 - Named for the presence of six legs.
 - All legs are uniramous.
 - Three tagmata: head, thorax, and abdomen.
 - Appendages on head and thorax; abdominal appendages reduced or absent.
- Classes:
 - Entognatha: Small group; bases of mouthparts enclosed within the head capsule (entognathous).
 - Orders: Protura, Diplura, Collembola (springtails).
 - Insecta: Enormous class; bases of mouthparts lie outside the head capsule (ectognathous).
 - Pterygotes (winged) and Apterygotes (wingless).

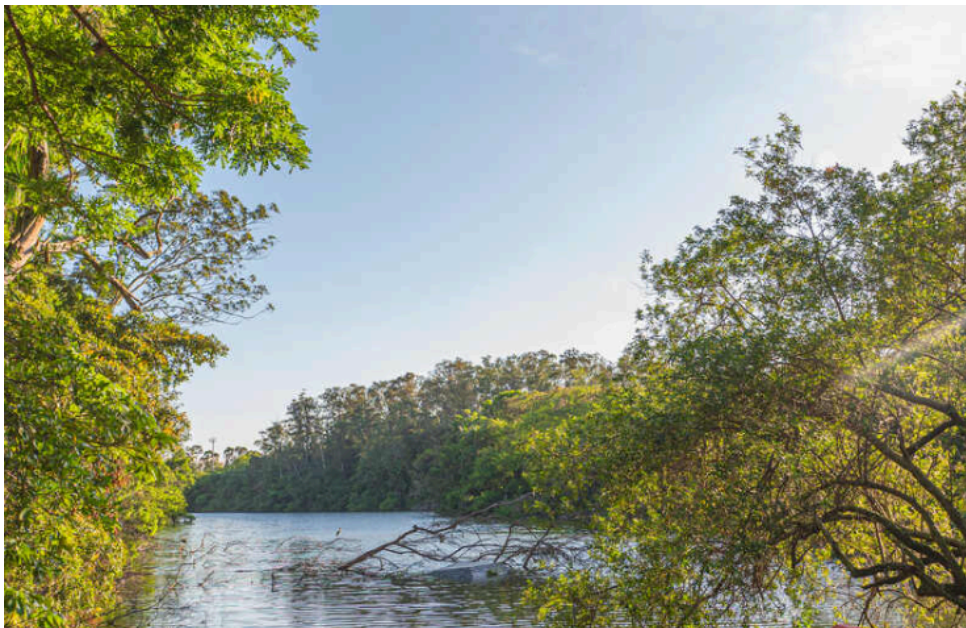


Figure 1: Cladogram showing hypothetical relationships among hexapods. Many synapomorphies have been omitted. Orders Protura, Collembola, and Diplura are entognathous. These orders, plus Thysanura, originated before the earliest winged ancestors. Orders Odonata and Ephemeroptera form Paleoptera, where wings are outspread. The remaining orders have wings that can fold back over the abdomen (Neoptera).

Class Insecta

- Diversity: Most diverse and abundant group of arthropods; more species than all other animals combined.
- Distribution: Widespread in all land habitats, freshwater, and some marine environments (e.g., **Halobates**).
- Adaptability:
 - Structural modifications in wings, legs, antennae, mouthparts, and alimentary canals.
 - Adapted to dry environments (hard exoskeleton limits evaporation).

External Form and Function

- Exoskeleton: Formed of complex plates called sclerites, connected by flexible hinge joints.
- Tagmata:

- Head: Bears compound eyes, antennae, and ocelli. Mouthparts (labrum, mandibles, maxillae, labium, hypopharynx).
- Thorax: Three segments (prothorax, mesothorax, metathorax). Bears 3 pairs of legs and usually 2 pairs of wings.
- Abdomen: 9-11 segments; bears cerci at posterior end; genitalia.



Figure 2: A, External features of a female grasshopper. The terminal segment of a male with external genitalia is shown in inset. B, Frontal view of head. C, A pair of lubber grasshoppers, **Romalea guttata** (order Orthoptera), copulating.



Figure 3: A few types of insect antennae.



Figure 4: A, A giant horned beetle, ***Diloboderus abderus*** (order Coleoptera), from Uruguay. Though the ferocious-looking processes from the head and thorax might appear to be for pinching or stabbing an opponent, they actually are used to lift or pry up a rival of the same species away from resources. B, Walking sticks, ***Diapheromera femorata*** (order Phasmatodea), mating. The species is common in much of North America. It is wingless, and despite its camouflage as a twig, it is eaten by numerous predators.

- Legs: Modified for various functions:
 - Walking (terminal pads and claws).
 - Jumping (hindlegs of grasshoppers).
 - Burrowing (mole crickets).
 - Swimming (water bugs, beetles).
 - Grasping prey (praying mantis).
 - Collecting pollen (honey bees).



Figure 5: Hindleg of grasshopper. Muscles that operate the leg are found within a hollow cylinder of exoskeleton. Here they are attached to the internal wall, from which they manipulate segments of limb on the principle of a lever. Note pivot joint and attachment of tendons of extensor and flexor muscles, which act reciprocally to extend and flex the limb.



Figure 6: A, Praying mantis (order Mantodea) feeding on an insect. B, Praying mantis laying eggs.



Figure 7: Adaptive legs of worker honey bee. In the foreleg, the toothed indentation covered with the velum combs out the antennae. The spur on the middle leg removes wax from wax glands on the abdomen. Pollen brushes on the front and middle legs comb off pollen picked up on body hairs and deposit it on the pollen brushes of the hindlegs. Long hairs of the pollen comb on the hindleg remove pollen from the comb of the opposite leg; then the auricle (pollen packer) presses it into a pollen basket when the leg joint is flexed back. A bee carries her load in both baskets to the hive and pushes pollen into a cell, to be cared for by other workers.

- Wings:
 - Cuticular extensions formed by the epidermis.
 - Double membrane with veins.
 - Variations: Thin/membranous (flies), thick/horny (beetles), parchmentlike (grasshoppers), scaly (butterflies/moths), hairy (caddisflies).
- Locomotion:
 - Walking: Tripod arrangement for stability.
 - Water Walking: Nonwetting hairs on footpads (e.g., water striders).



Figure 8: Water strider, **Gerris** sp. (order Hemiptera). The animal is supported on its long, slender legs by the water's surface tension.

- Flight:
 - Only invertebrates capable of flight.
 - Direct flight muscles: Attached to the wing itself.
 - Indirect flight muscles: Attached to the thorax (alter shape of thorax to move wings).
 - Neural Control:
 - Synchronous: Single nerve impulse per muscle contraction (e.g., butterflies).
 - Asynchronous: Occasional nerve impulses keep muscles contracting/relaxing; allows rapid wing beats (e.g., flies, bees).
 - Wing movement: Figure-eight pattern for forward thrust and lift.

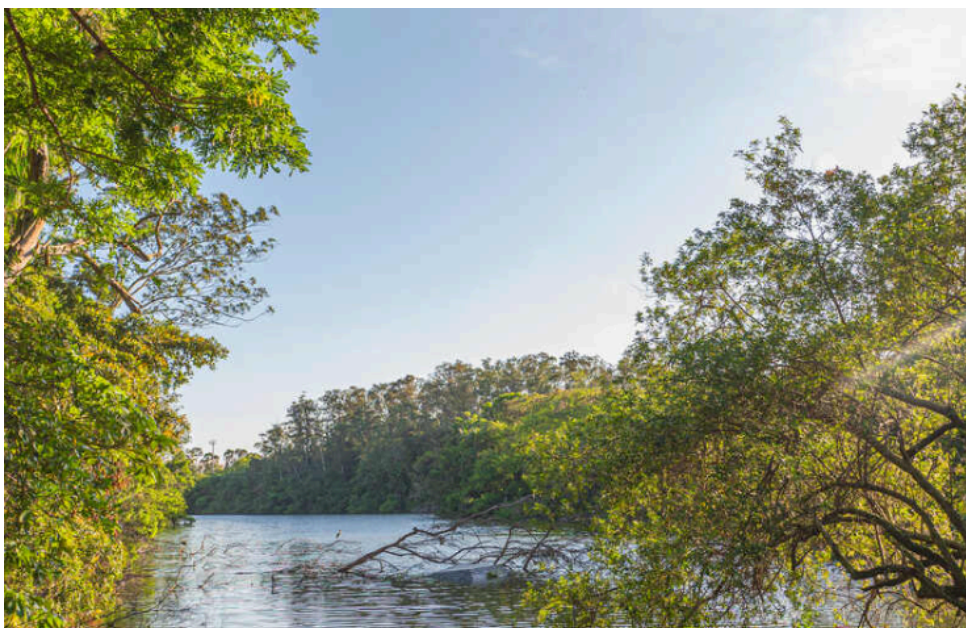


Figure 9: House fly, **Musca domestica** (order Diptera). House flies can become contaminated with over 100 human pathogens, which may be transferred to human and animal food by direct contact, regurgitated food, and feces.



Figure 10: A, Flight muscles of insects such as cockroaches, in which upstroke is by indirect muscles and downstroke is by direct muscles. B, In insects such as flies and bees, both upstroke and downstroke are by indirect muscles. C, The figure-eight path followed by the wing of a flying insect during the upstroke and downstroke.

Internal Form and Function

- Nutrition:
 - Digestive System: Foregut (mouth, esophagus, crop, gizzard), Midgut (stomach, gastric ceca), Hindgut (intestine, rectum, anus).
 - Feeding Habits: Phytophagous (herbivores), Saprophagous (dead animals), Predaceous, Parasitic (including Hyperparasitism and Parasitoids).



Figure 11: Internal structure of female grasshopper.



Figure 12: Female human flea, **Pulex irritans** (order Siphonaptera).



Figure 13: **Gliricola porcelli** (order Phthiraptera), a chewing louse of guinea pigs. Antennae are normally held in the deep grooves on the sides of the head.



Figure 14: The head and body louse of humans, **Pediculus humanus** (order Phthiraptera), feeding.



Figure 15: A, Larval stage of the tomato hornworm, **Manduca sexta** (order Lepidoptera). The more than 100 species of North American sphinx moths are strong fliers and mostly nocturnal feeders.

Their larvae are called hornworms because of the large, fleshy posterior spine. B, Hornworm parasitized by a tiny wasp, **Apanteles** (a parasitoid), which laid its eggs inside the caterpillar. The wasp larvae have emerged, and their pupae are formed on the caterpillar's skin. Young wasps emerge in 5 to 10 days, but the caterpillar dies.

- Mouthparts:
 - Sucking: Tube for piercing (e.g., mosquitoes) or coiled proboscis (e.g., butterflies).
 - Sponging/Lapping: Soft lobes for liquid food (e.g., house flies).
 - Chewing: Strong mandibles for crushing (e.g., grasshoppers).



Figure 16: Four types of insect mouthparts. A, Chewing mouthparts of a grasshopper. B and C, Sucking mouthparts of a mosquito. Parts of piercing fascicle are shown in cross section (C). D, Sucking mouthparts of a butterfly. Mandibles are absent, and maxillae form a long proboscis. E, Sponging mouthparts of a house fly. A pair of large lobes with grooves on their lower surface are on the end of the labium.

- Circulation: Tubular heart, dorsal aorta, hemolymph (usually no oxygen transport function).
- Gas Exchange:
 - Tracheal System: Network of thin-walled tubes (tracheae) branching into tracheoles.
 - Spiracles: External openings with valves.
 - Aquatic Respiration: Diffusion, tracheal gills, or air bubbles (plastron).



Figure 17: A, Generalized arrangement of insect tracheal system (diagrammatic). Air sacs and tracheoles not shown. B, Relationship of spiracle, tracheae, taenidia (chitinous bands that strengthen the tracheae), and tracheoles (diagrammatic).

- Excretion and Water Balance:

- Malpighian Tubules: Blind tubules attached to midgut/hindgut juncture.
- Function: Active transport of ions (H^+ , K^+), water follows by osmosis. Uric acid precipitates in hindgut, water/salts reabsorbed in rectum.
- Efficient water conservation.



Figure 18: Malpighian tubules of insect. A, Malpighian tubules are located at the juncture of the midgut and hindgut (rectum) as shown in the cutaway view of a wasp. B, Function of Malpighian tubules. Hydrogen ions are actively exchanged for potassium ions in the upper tubules. Water and potassium acid urate (KHUr) follow. Potassium is resorbed with water and other solutes in the rectum.

- Nervous System: Fused ganglia, giant fiber system, stomodeal nervous system.
- Sense Organs:
 - Mechanoreception: Sensilla (setae).
 - Auditory: Tympanal organs (e.g., in Orthoptera).
 - Chemoreception: Taste and smell (mouthparts, antennae, tarsi).
 - Visual:
 - Simple eyes (ocelli): Monitor light intensity.
 - Compound eyes: Composed of ommatidia; high flicker-fusion frequency; color vision (often UV sensitive).



Figure 19: Compound eye of an insect. A single ommatidium is shown enlarged to the right.

- Reproduction:
 - Sexes separate; internal fertilization.
 - Sperm transfer: Direct or via spermatophores.
 - Oviposition: Specialized sites (specific plants, water, hosts).



Figure 20: Copulation in insects (see also Figures 21.2B and 21.7B). A, **Omura congrua** (order Orthoptera) are a kind of grasshopper found in Brazil. B, Bluet damselflies, **Enallagma** sp. (order Odonata), are common throughout North America. Here, a male still grasps a female after copulation. The female (white abdomen) lays eggs in the water.



Figure 21: An ichneumon wasp with the end of the abdomen raised to thrust her long ovipositor into wood to find a tunnel made by the larva of a wood wasp or wood-boring beetle. She can bore 13 mm or more into the wood to lay her eggs in the larva of a wood-boring beetle, which will become host for the ichneumon larvae. Other ichneumon species attack spiders, moths, flies, crickets, caterpillars, and other insects.

Metamorphosis and Growth

- Instar: Stage between molts.
- Ametabolous (Direct) Development:
 - Young similar to adults except in size/sexual maturity.
 - Primitively wingless insects (e.g., silverfish).



Figure 22: Silverfish **Lepisma** (order Thysanura) is often found in homes.

- Hemimetabolous (Incomplete) Metamorphosis:
 - Stages: Egg -> Nymph -> Adult.
 - Nymphs have external wing buds.

- Aquatic nymphs often have gills.
- Examples: Grasshoppers, dragonflies, mayflies.



Figure 23: A, Ecdysis in a cicada, **Tibicen davisi** (order Hemiptera). The old cuticle splits along a dorsal midline as a result of increased blood pressure and of air forced into the thorax by muscle contraction. The emerging insect is pale, and its new cuticle is soft. The wings will be expanded by blood pumped into veins, and the insect enlarges by taking in air. B, An adult **Tibicen davisi**.

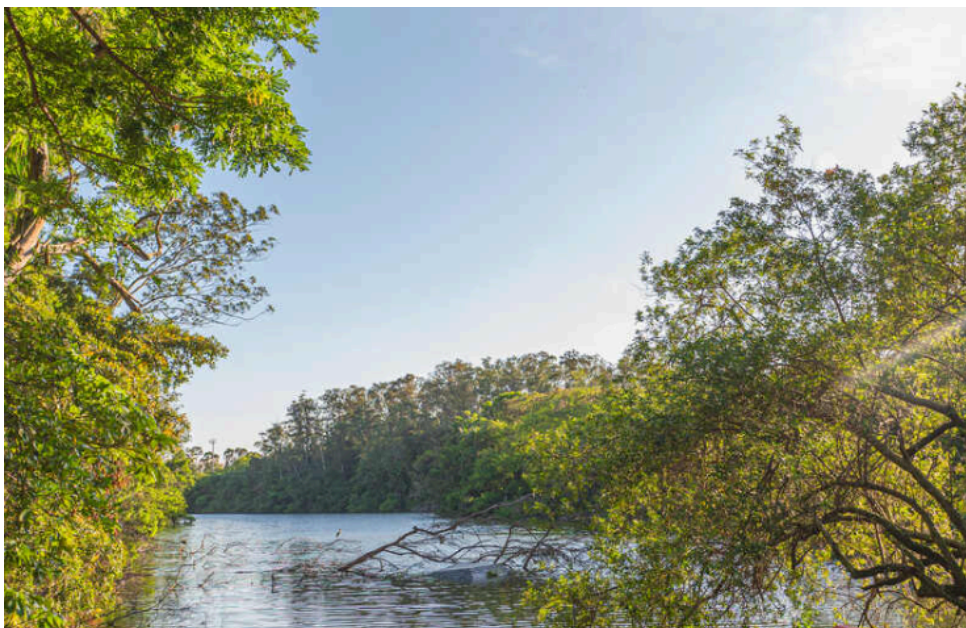


Figure 24: Life history of a hemimetabolous insect.



Figure 25: A, Stonefly, **Perla** sp. (order Plecoptera). B, Ten-spot dragonfly, **Libellula pulchella** (order Odonata). C, Nymph of dragonfly. Both stoneflies and dragonflies have aquatic nymphs that undergo gradual metamorphosis.

- Holometabolous (Complete) Metamorphosis:
 - Stages: Egg -> Larva -> Pupa -> Adult.
 - Larvae (caterpillars, maggots, grubs) very different from adults.
 - Pupa is a nonfeeding, transitional stage.
 - 88% of insects.



Figure 26: Complete (holometabolous) metamorphosis in a butterfly, **Danaus plexippus**. Eggs hatch to produce first of several larval instars. Last larval instar molts to become a pupa. Adult emerges at pupal molt.



Figure 27: A, Mosquito **Culex** (order Diptera) lays her eggs in small packets or rafts on the surface of standing or slowly moving water. B, Mosquito larvae are the familiar wrigglers of ponds and ditches.

To breathe, they hang head down, with respiratory tubes projecting through the surface film of water. Motion of vibratile tufts of fine hairs on the head brings a constant supply of food to these filter feeders.

- Physiology of Metamorphosis:
 - Hormones: PTTH (brain) -> Ecdysone (prothoracic gland) -> Molting.
 - Juvenile Hormone (corpora allata): Determines outcome of molt (Larva -> Larva if high; Larva -> Pupa if low; Pupa -> Adult if absent).
- Diapause: Period of dormancy (hibernation/estivation) triggered by environmental cues (photoperiod).

Defense

- Coloration:
 - Mimicry: Batesian (palatable mimics unpalatable) or Müllerian.
 - Aposematic Coloration: Warning colors.
 - Crypsis: Camouflage.
- Chemical Warfare: Repulsive odors, sprays, bad tastes.



Figure 28: Mimicry in butterflies. A, Monarch butterfly is distasteful to, and avoided by some birds because as a caterpillar it fed on the acrid milkweed. B, The monarch is mimicked by the smaller viceroy butterfly, **Limenitis archippus**, which feeds on willows and may (or may not) be tasteful to birds, but is not eaten because it so closely resembles the monarch in color and markings. This kind of mimicry is called Batesian mimicry, although recent evidence argues for Müllerian mimicry.

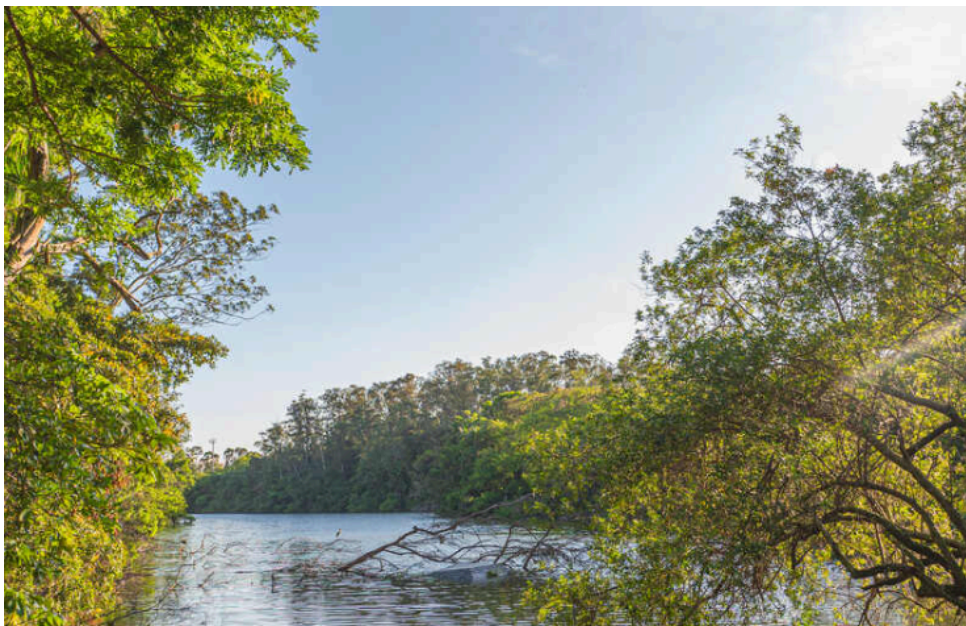


Figure 29: Crypsis (camouflage) in insects. A, **Estigena pardalis** (order Lepidoptera) in Java resembles a dead leaf. B, Bizarre processes from the thorax of a treehopper from Mexico, **Sphongophorus** sp. (order Hemiptera), masquerade as parts of the twig on which it feeds. C, Broken outlines and color of a katydid (**Dysonia** sp., order Orthoptera) in Costa Rica give it the appearance of the leaves on which it has been feeding.

Behavior and Communication

- Behavior: Innate and learned behaviors (e.g., potter wasp, dung beetles).



Figure 30: Dung beetles **Canthon pilularis** (order Coleoptera), chew off a bit of dung, form it into a ball, and then roll it to where they will bury it in soil. One beetle pushes while the other pulls. Eggs are laid in the ball, and larvae feed on the dung. Dung beetles are black, an inch or less in length, and common in pasture fields.

- Communication:
 - Chemical: Pheromones (attraction, alarm, trail marking, caste determination).
 - Sound: Warning, courtship (e.g., crickets, cicadas).
 - Tactile: Tapping, stroking.
 - Visual: Bioluminescence (e.g., fireflies).

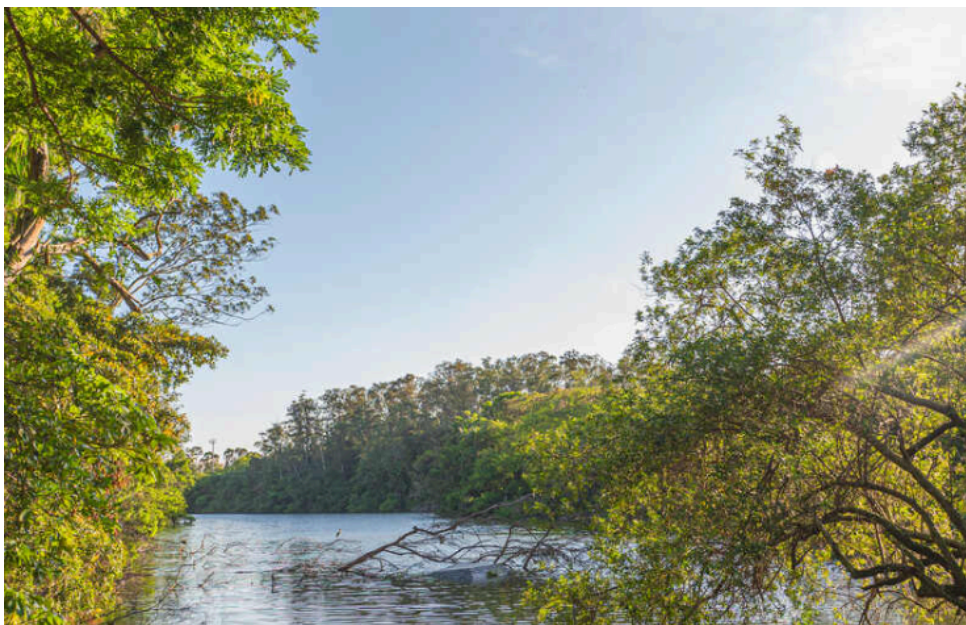


Figure 31: Firefly femme fatale, **Photuris versicolor**, eating a male **Photinus tanytoxus**, which she has attracted with false mating signals.

- Social Behavior:
 - Eusociality: Cooperative brood care, overlapping generations, caste system.

- Honey Bees: Queen (fertile female), Drones (males), Workers (sterile females). Communication via dance.
- Termites: King and Queen, Workers, Soldiers. Trophallaxis.
- Ants: Queen, Workers, Soldiers. Complex behaviors (farming, slavery).



Figure 32: Queen bee surrounded by her court. The queen is the only egg layer in the colony. The attendants, attracted by her pheromones, constantly lick her body. As food and the queen's pheromones are transferred from these bees to others, the queen's presence is communicated throughout the colony.



Figure 33: A, Termite workers, **Reticulitermes flavipes** (order Isoptera), eating yellow pine. Workers are wingless immature animals that tend the nest. B, Termite queen becomes a distended egg-laying machine. The queen and several workers and soldiers are shown here.



Figure 34: A, An ant (order Hymenoptera) tending a group of aphids (order Homoptera). The aphids feed copiously on plant juices and excrete the excess as a clear liquid rich in carbohydrates (“honeydew”), which is cherished as a food by ants. B, A weaver ant nest in Australia.

Insects and Human Welfare

- Beneficial Insects:
 - Pollination (coevolution with plants).
 - Products (honey, silk, shellac).
 - Predators/Parasitoids of pests.

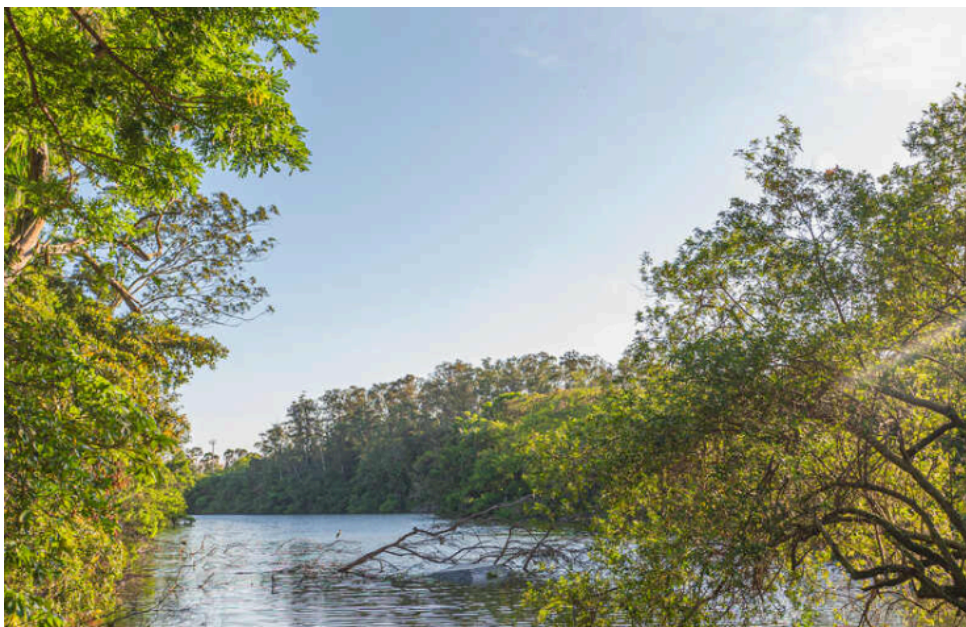


Figure 35: Some beneficial insects. A, A predaceous stink bug (order Hemiptera) feeds on a caterpillar. Note the sucking proboscis of the bug. B, A ladybug beetle (order Coleoptera). Adults (and larvae of most species) feed voraciously on plant pests such as mites, aphids, scale insects, and thrips. C, A parasitic wasp (***Larra bicolor***) attacking a mole cricket. The wasp drives the cricket from its burrow, then stings and paralyzes it. After the wasp deposits her eggs, the mole cricket recovers and resumes an active life—until it is killed by developing wasp larvae.

- Harmful Insects:
 - Crop pests (defoliators, borers, suckers).
 - Disease vectors (Mosquitoes - Malaria, West Nile; Fleas - Plague; Lice - Typhus).
 - Property destruction (termites, clothes moths).



Figure 36: Some insect pests. A, Japanese beetles, **Popillia japonica** (order Coleoptera), are serious pests of fruit trees and ornamental shrubs. They were introduced into the United States from Japan in 1917. B, Longtailed mealybug, **Pseudococcus longispinus** (order Hemiptera). Many mealybugs are pests of commercially valuable plants. C, Corn ear worms, **Heliothis zea** (order Lepidoptera). An even more serious pest of corn is the infamous corn borer, an import from Europe in 1908 or 1909.



Figure 37: A mosquito, **Anopheles quadrimaculatus** (order Diptera). **Anopheles** spp. are vectors of malaria.

- Control:
 - Insecticides (issues: resistance, environmental persistence).

- Biological Control (pathogens, predators, sterile males).
- Integrated Pest Management (IPM).

Classification and Phylogeny

- Orders: (Examples shown in figures)
 - Ephemeroptera (Mayflies), Odonata (Dragonflies), Orthoptera (Grasshoppers), Hemiptera (Bugs), Coleoptera (Beetles), Lepidoptera (Butterflies/Moths), Diptera (Flies), Hymenoptera (Bees/Wasps/Ants), etc.



Figure 38: Mayfly (order Ephemeroptera). A, Nymph. B, Adult.



Figure 39: Oak treehoppers, **Platycotis vittata** (order Hemiptera).



Figure 40: Adult ant lion (order Neuroptera).



Figure 41: A, **Papilio krishna** (order Lepidoptera) is a beautiful swallowtail butterfly from India. Members of the Papilionidae grace many areas of the world, both tropical and temperate, including North America. Compare knobbed antennae with plumed antennae in B, **Rothschildia jacobaea**, a saturniid moth from Brazil. **Hyalophora cecropia** is a common saturniid in North America. C, Paper wasps (order Hymenoptera) attending their pupae and larvae. D, **Curculio proboscideus**, the chestnut weevil, is a member of the largest family (Curculionidae) of the largest insect order (Coleoptera). This family includes many serious agricultural pests.

- Phylogeny:
 - Hexapods + Crustaceans = Pancrustacea.
 - Evolution of wings: Possibly from thoracic gills of aquatic ancestors.



Figure 42: A, A Paleozoic “mayfly” nymph with thoracic winglets and abdominal gills. The thoracic winglets could have been precursors of wings. B, A Paleozoic insect with multiramous thoracic legs and vestigial, multiramous abdominal limbs.



Figure 43: An ancient paleopterous insect, **Homolaneura joannae**, from the Upper Carboniferous illustrating two functional pairs of wings as well as a pair of prothoracic winglets. The prothoracic winglets are broadly attached and hinged like the mesothoracic and metathoracic wings of some recent insects. Prior to the development of fully functional wings, hinged and mobile thoracic winglets on all three thoracic segments may have served a thermoregulatory purpose that eventually allowed the evolution of winglets large enough to support flight.